Enhanced Algorithm for Water Transparency Estimation in Turbid Plateau Waters Using Orbita Hyperspectral (OHS) Imagery

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***Abstract*— Deteriorating water environments in plateau lakes are increasingly influenced by climate change and human activities. Water transparency, critical for understanding underwater lightfield environments, is commonly quantified as the Secchi disk depth (Z_{SD}, m). Despite advances in Z_{SD} semianalytical model, their application in turbid plateau lakes faces challenges due to differences in quasi-analytical algorithm (QAA) and image limitations. To address these challenges, this study introduced a novel hybrid QAA model (QAA_{hybrid}) specifically designed to estimate Z_{SD} using orbita hyperspectral (OHS) images, a new hyperspectral image in China. The algorithm's uncertainty and image quality were evaluated and compared using the error propagation theory and noise equivalent Z_{SD}(NE_{ZSD}). Several main findings can be drawn: 1) the QAA_{hybrid}, categorized as moderately turbid and extremely turbid waters using a remote sensing reflectance ratio and outperformed other QAA models; 2) the new Z_{SD} model produced a mean absolute percentage difference (MAPD) of 10.89%, demonstrating better accuracy compared to the existing Z_{SD} models and had an MAPD of 23.35% when applied to OHS images; and 3) documented Z_{SD} from OHS images showed that Dianchi Lake had a trend of increasing from the lake center toward the shore, while Erhai Lake had a trend of decreasing from north to south. These findings emphasize the feasibility of the new Z_{SD} semianalytical model and OHS data in water quality monitoring, providing a reliable approach for water environmental management.**

***Index Terms*— Hyperspectral remote sensing, semianalytical model, spatial–temporal patterns, uncertainty assessment, water transparency.**

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I. Introduction

INLAND waters are crucial ecosystems that serve as essential freshwater sources for industry, agriculture, and city dwellers [1]. However, water quality in these systems has degraded significantly in recent years, primarily due to human activities and climate change, which has emerged as a global concern [2]. Therefore, monitoring and assessing the health of inland water is critical for maintaining and managing aquatic environments [3].

Water transparency, as a visual indicator of the water's overall condition, directly affects human visual perception of the aquatic environment and regulates the availability of light in underwater ecosystems [4], [5]. Water transparency is typically measured using a Secchi disk (Z_{SD}, m), with the depth at which it becomes invisible in the water defined as the Secchi disk depth [6], [7]. With the advancement of satellite technology, remote sensing has offered a feasible way to achieve long-term and long-scale observations [8], [9], [10], [11]. Satellite imagery can document the spatial–temporal distribution of Z_{SD}, but small lake area and complex bio-optical features limit Z_{SD} monitoring [12], [13], [14]. Over recent decades, various types of coarse satellite images [e.g., Moderate Resolution Imaging Spectroradiometer (MODIS), Ocean and Land Color Instrument (OLCI), and Geostationary Ocean Color Imager (GOCI)] were mainly applied to quantify Z_{SD},

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but the low spatial resolution may lack sensitivity to the local Z_{SD} spatial patterns [15], [16], [17]. Elaborate satellite imagery [e.g., operational land imager (OLI) and multispectral instrument (MSI)] has been employed for Z_{SD} retrieval, but the broad bandwidth makes it difficult to accurately capture spectral absorption and scattering features, which affects the accuracy of Z_{SD} retrieval [18], [19], [20]. The launch (April 2018) of the orbita hyperspectral (OHS) satellite addresses these concerns by providing high resolutions (2.5-D temporal resolution, 2.5-nm spectral resolution, and 10-*m* spatial resolution) complementary metal–oxide–semiconductor (CMOS) image product. OHS imagery holds significant potential for observing inland water quality, especially for those with bays, estuaries, reservoirs, and small lakes [21], [22]. OHS images have been successfully used to monitor chlorophyll-a and heavy metals in specific areas [23], [24], but the feasibility of monitoring Z_{SD} in various waters needs further verification.

There are two main approaches to estimating Z_{SD} from remote sensing data: the empirical model and semianalytical model [25]. The empirical model establishes a regression relationship between measured Z_{SD} and remote sensing data but lacks generalizability and is specific to certain regions [26], [27], [28]. Conversely, the semianalytical model retrieval Z_{SD} is derived from the new underwater visibility theory [18], [29], [30], has demonstrated promising performance, and extensively deployed in various environments and remote sensing platforms [4], [8], [18], [31], [32], [33]. For the new underwater visibility theory, the accuracy of the quasi-analytical algorithm (QAA) plays a critical role in the calculation of the diffuse attenuation coefficient (K_d); however, previous studies have shown that the original QAA (QAA_{V6}) might be unsuitable for complex inland waters [33], [34]. First, the uncertainties in the QAA estimates are propagated to the underwater visibility theory, affecting Z_{SD} estimates [4]. In addition, the variable water conditions and optical environments are challenging for the application of QAA [34], [35]. For example, the QAA_{Y22} model developed for Taihu Lake performs poorly in Dianchi Lake [36]; while the QAA_{Y13} model, which used in situ data from Dianchi Lake, also produced unsatisfactory results due to its long reference wavelength [37]. Water classification facilitates the understanding of variability among these waters, providing insights for wider model applications [38]. Incorporating different reference bands significantly enhances the retrieval accuracy of the QAA models [36]. Therefore, it is necessary to improve the QAA model and develop a new semianalytical model of Z_{SD} that can be widely applied for the complexity of turbid inland waters.

The complexity of inland waters motivates the exploration of OHS imagery capabilities for water environmental management and the advancement of the new Z_{SD} model. Consequently, the following objectives were specified to: 1) improve a novel QAA model (QAA_{hybrid}) suitable for various waters and construct a new Z_{SD} semianalytical model based on underwater visibility theory; 2) assess the estimation performance of OHS for Z_{SD} in inland turbid waters (uncertainty and noise equivalent Z_{SD}); and 3) analyze the spatial–temporal patterns of OHS-documented Z_{SD}.

II. Data and Method

A. Overview of the Study Area

Tow turbid plateau lakes (Fig. 1), Dianchi Lake (DCL; 330 km^2, 102°37′ E∼102°48′ E, and 24°40′ N∼25°02′ N) and Erhai Lake (EHL; 252 km^2, 100°08′ E∼100°31′ E, and 25°61′ N∼25°98′ N), are in the Yunnan–Guizhou Plateau. The DCL is marked by severe eutrophication and high turbidity, primarily due to increased concentration of suspended sediments and chlorophyll-a [39], [40]. Similarly, the EHL is experiencing a gradual increase in pollution, exhibiting moderate eutrophication and turbidity [41], [42].

Plateau characteristics further complicate remote sensing monitoring, with intense light conditions affecting the optical response of water and pronounced atmospheric status interfering with image analysis [43], [44]. Whereas current QAA models are usually based on idealized water conditions that are unable to accurately reflect the effect of turbidity on remote sensing signals [45]. To incorporating optical conditions specific to turbid plateau water to accurately assess Z_{SD} changes. To accurately evaluate Z_{SD} variations in plateau turbid waters, it is essential to develop a new semianalytical model tailored to the optical properties of these waters. Furthermore, a comprehensive comparison of various atmospheric correction (AC) models is essential to reduce the atmospheric interference and enhance the reliability of the assessment, ensuring that results are more suited to the distinct characteristics of the region.

B. In Situ Data and Processing

From April 2017 to November 2017, four field campaigns were conducted in DCL and EHL to measure and collect Secchi-disk depth (Z_{SD}), water samples, and remote sensing reflectance (R_{rs}, sr^{-1}), with a total of 107 sampling sites obtained (Fig. 1, 70 samples in DCL and 37 samples in EHL). To reduce measurement error in Z_{SD}, a standard 30-cm diameter Secchi disk was used by the same operator to measure Z_{SD} at each site, avoiding measurements in the waves, winds, and shaded side of the boat [1]. Simultaneously, measurements of the upward radiance (L_u), downward sky radiance (L_{sky0}), and radiance of the reference plaque (L_p), were quantified using an analytical spectral devices (ASDs) Inc. FieldSpec Pro spectrometer (spectral range extending from 350 to 1050 nm with measurements recorded at 1.5-nm increments across 512 channels) following above-water method [46], [47]. Subsequently, R_{rs} was calculated as follows [48]:

$$R_{rs} = R_g\left(L_u - \rho L_{sky}\right)/L_p\pi \tag{1}$$

where ρ represents the skylight reflectance on the water surface, varying with wind conditions (0.022 for calm weather, 0.025 for 5 m/s, and 0.026–0.028 for 10 m/s), and R_g indicates the reflectance for the gray plaque (0.3) [49].

The optical parameters, including the absorption of chromophoric-dissolved organic matter (a_{CDOM}), nonalgal particle (a_{nap}), and phytoplankton (a_{ph}) were measured following the quantitative filtering technique [50]. The measurements and the absorption coefficient of pure water (a_w) were used

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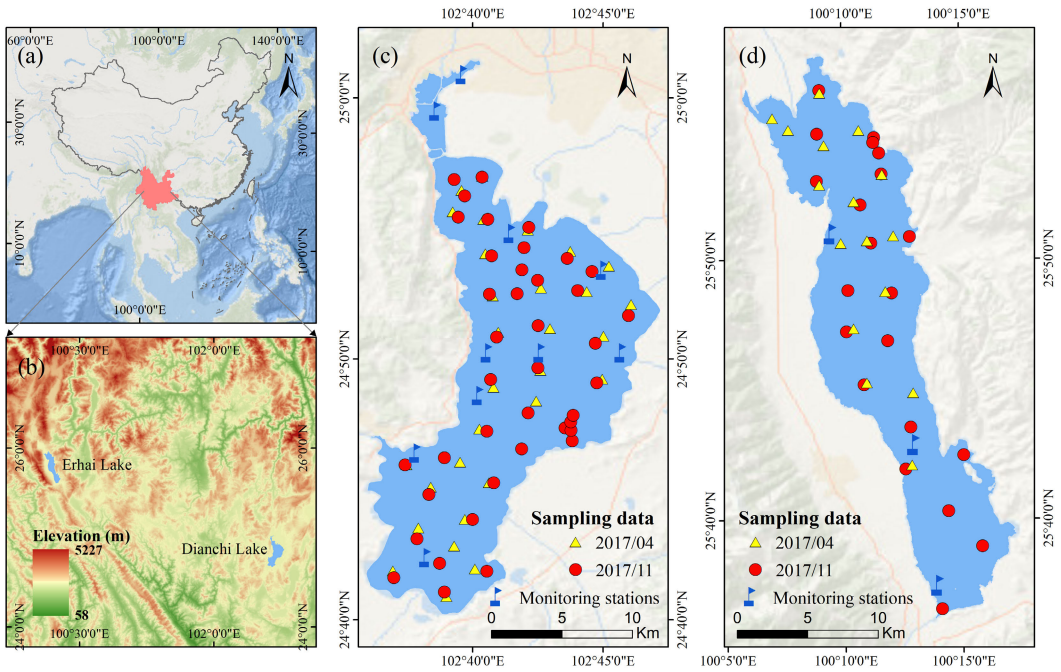


Fig. 1. Study lakes. (a) Study area in China, (b) geographical distribution of lakes, (c) and (d) spatial location of sampling data in Dianchi Lake and Erhai Lake.

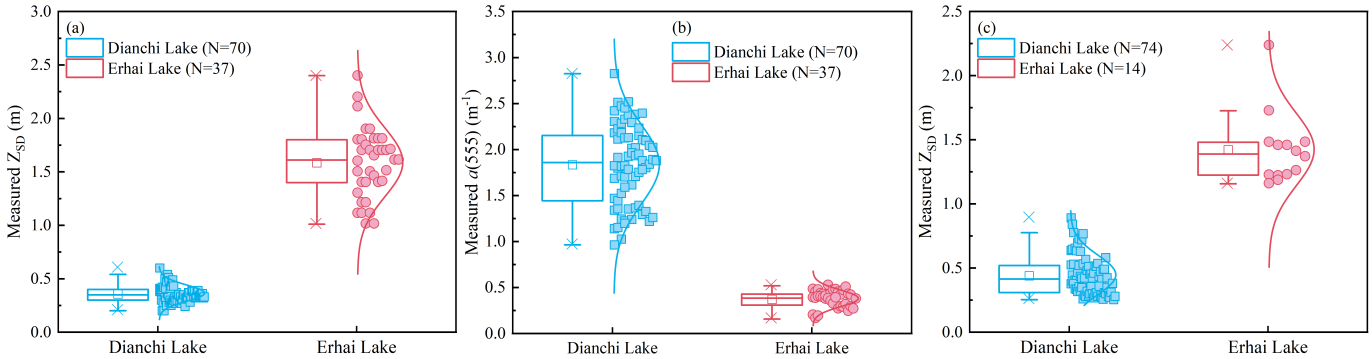


Fig. 2. Data used for model development, validation, and satellite synchronous verification, including in situ Z_{SD} and $a(555)$. (a) Semianalytical model development. (b) QAA_{hybrid} model development. (c) Image validation.

to derive the total absorption coefficient (a). These data were employed to develop the QAA model.

Furthermore, field measurements acquired Z_{SD} data that were synchronized with OHS observations provided by the China National Environmental Monitoring Centre (Fig. 1). In DCL, there were 74 matchup samplings (from November 2020 to September 2021), while in EHL, there were 14 matchup samplings (from November 2020 to Jan 2022). These matched Z_{SD} were used to validate the OHS-derived Z_{SD}, while measured Z_{SD} were used for developing the semi-analytical model (Fig. 2).

C. Satellite Data and Processing

The OHS imagery data (Level-1A) were acquired from Zhuhai Orbita Aerospace Technology Company Ltd. (https://www.obtdata.com/). A total of 11 cloudless images covering the DCL from April 2019 to September 2021, and 16 cloudless images covering the EHL from October 2018 to January 2022 were acquired. The OHS imagery has 256 spectral bands (optional) spanning a range of 400–1000 nm and 32 bands with a bandwidth range of 5–14 nm, which can be imaged as required.

The processing of OHS images included radiometric calibration, AC, and orthorectification correction. Although many studies have shown that the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) model was suitable for OHS images [23], [51], further probing was needed for the estimation of Z_{SD}. In this study, the dark object subtraction (DOS) model, the quick atmospheric correction (QUAC) model, the second simulation of the satellite signal in the solar spectrum (6S) model, and the FLAASH model were compared [52], [53]. The process of AC models was described by Zheng et al. [23]. After radiometric calibration, four AC models were used and compared to correct for atmospheric effect. The OHS images had a geometrical deviation (<1 km), and these were corrected for deviation using OPENOHS [23], a software designed for processing the OHS image (https://www.obtdata.com/#/index).

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After processing the imagery, the lake boundaries were obtained from the GLAKES dataset. To reduce the influence of proximity effects, the boundaries was reduced inward by 3 pixels [54], [55]. Subsequently, water pixels within the boundaries were extracted using the normalized difference water index (NDWI), with a threshold value of 0.48 [56], [57].

D. Accuracy and Uncertainty Assessment

Generally, the algorithm accuracy is statistically validated by comparing the estimated values with in situ values [58]. However, such accuracy metrics do not sufficiently convey the uncertainties associated with satellite data or the algorithm retrieval processes [59]. Previous studies have focused on improving the accuracy of the QAA model, the explanation of errors throughout the entire "QAA model, semianalytical model, and image application" process remains relatively underexplored [19], [60], [61].

To improve the understanding of this process, this study conducted three targeted evaluations.

1) Evaluation of the QAA reference band, error propagation theory can evaluate the uncertainties propagate from the reference band to the extrapolated band within the QAA model [59].
2) The sensitivity evaluation of the semianalytical model, random errors were systematically introduced into the key parameters of the semianalytical model (ensuring consistency, these random errors were set as fixed values rather than randomly generated within a range).
3) Quality evaluation of OHS image, noise equivalent transparency (NE_{ZSD}) can quantitatively describe the algorithm's performance within the imagery.

The statistical accuracy was evaluated using conventional metrics, such as the coefficient of determination (R^2), the statistical significance (p-value), the root mean square difference (RMSD), and the mean absolute percentage difference (MAPD) [62]

$$\mathrm{MAPD} = \frac{1}{n}\sum_{i,j=1}^{n}\left|\frac{x_i - y_j}{x_i}\right| \times 100\% \tag{2}$$

$$\mathrm{RMSD} = \sqrt{\frac{1}{n}\sum_{i,j=1}^{n}\left(x_i - y_j\right)^2} \tag{3}$$

where x_i and y_j represent the measured and estimated values and n represents the number of samples.

III. Model Development

Estimating Z_{SD} using the semianalytical model includes four steps: 1) classifying various waters using a R_{rs} ratio; 2) calculating a and backscattering of particulate (b_{bp}) with different QAA models; 3) obtaining K_d from a and b_{bp} for various waters; and 4) calibrating the consistency of the Z_{SD} model. The flowchart is shown in Fig. 3, and the detailed step is exhibited in Table I.

A. Water Classification

Previous studies have shown that a single QAA model lacks extensibility, demonstrating the need to use different QAA models for different waters [36], [45], [60]. The detailed single QAA model results can be found in Fig. S1. Referring to Xiang et al. [63] and the band setting of the OHS image, an R_{rs} ratio [$f = R_{rs}(670)/R_{rs}(500)$] was employed for water classification. This ratio effectively distinguishes the turbidity of water by comparing the reflection peak of algae-dominated water at the red band with the strong scattering signal of suspended solids in blue band, while the DCL and EHL have different levels of chlorophyll-a and suspended solids [38]. Fig. 4 illustrates the relationship between f and Z_{SD}, which are two water types: moderately turbid (low f values and high Z_{SD}) and extremely turbid (high f values and low Z_{SD}) [63]. In this study, $f = 0.70$ was used to separate extremely turbid and moderately turbid waters [63]. A comparison of the different water classification models is detailed in Fig. S2.

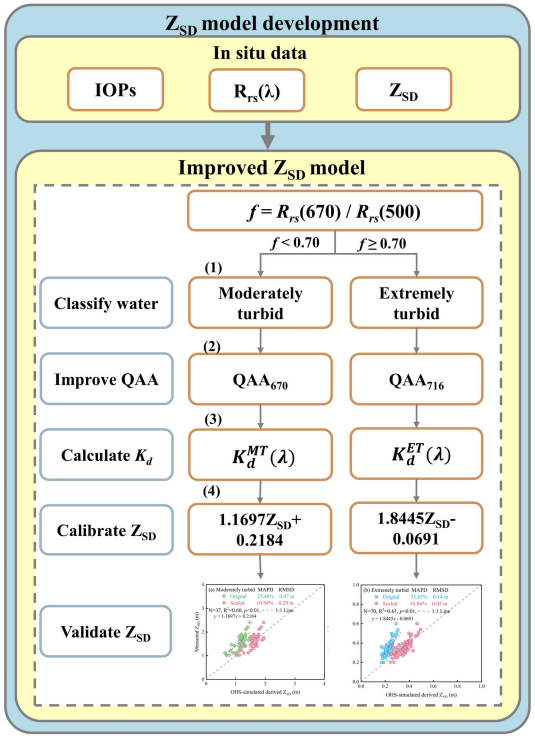


Fig. 3. Flowchart of the Z_{SD} semianalytical model.

B. Improving the QAA Model

The QAA model was introduced to derive a and b_{bp} at different wavelengths by R_{rs} [64]. This approach comprises two empirical algorithms following the oceanic conditions [65], which has indicated the necessity of adjusting two empirical algorithms for turbid waters [30], [34], [66]. The following are the improved steps for QAA_{hybrid}.

1) For moderately turbid waters ($f < 0.70$), 670 nm was continued to choose as the reference band, and selecting a different band did not yield reliable outcomes (Fig. S3). In addition, the parameter $a(670)$ was reparametrized using data from moderately turbid waters

$$a(\lambda_0) = a_w(670) + 0.7049\left(\frac{r_{rs}(670)}{r_{rs}(466) + r_{rs}(500)}\right)^{0.64} \tag{4}$$

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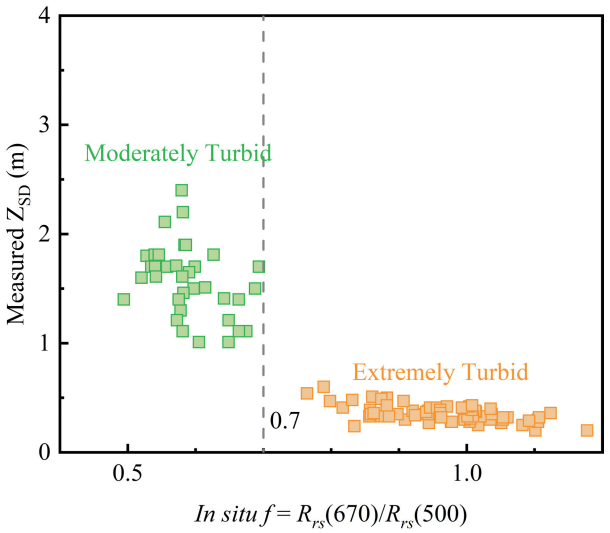


Fig. 4. Scatter plot between Z_{SD} and $f = R_{rs}(670)/R_{rs}(500)$.

where $r_{rs}(\lambda)$ is the R_{rs} under water. In the OHS image, the absence of the 443 and 490 nm bands required comparison and validation of different bands [Fig. 5(a)]. As a substitute, the bands at 466 and 500 nm were used. Similarly, for the particle backscattering coefficient (η), a 466 nm was employed as an alternative and compared various η in Fig. 6 [37], [61], [67]

$$\eta = 2.0 \times \left[1 - 1.2 \times e^{\left(-0.9 \times \frac{r_{rs}(466)}{r_{rs}(670)}\right)}\right]. \quad (5)$$

2) For extremely turbid waters ($f \geq 0.70$), 716 nm was selected as the reference band for two reasons [23]. First, $a_w(716)$ dominates $a(716)$ and the $a_{CDOM}(716)$ was negligible compared with $a_w(716)$. Second, 716 nm represents the center wavelength of the OHS image

$$a(\lambda_0) = a_w(716) + \alpha\frac{r_{rs}(550)}{r_{rs}(700)} + \beta\frac{r_{rs}(670)}{r_{rs}(700)} + \gamma\frac{r_{rs}(760)}{r_{rs}(550)}$$
$$\alpha = -0.122, \beta = 0.571, \gamma = 1.177 \quad (6)$$

where $\Delta a(\lambda)$ represents the contributions of the non-water constituents. The function for $a(716)$ was taken from the methodology introduced by Zheng et al. [23] and Rodrigues et al. [67], adjusted to accommodate the band settings of the OHS. The coefficients required for extremely turbid water data were computed accordingly [Fig. 5(b)].

Similarly, the function η was adapted for extremely turbid waters. The value of 2 in η has been changed to 2.2, following the findings of Rodrigues et al. [67] and comparing various η in Fig. 7

$$\eta = 2.2 \times \left[1 - 1.2 \times e^{\left(-0.9 \times \frac{r_{rs}(466)}{r_{rs}(716)}\right)}\right]. \quad (7)$$

C. Z_{SD} Model

Estimating Z_{SD} using the semianalytical model needs to obtain K_d from a and b_{bp}. The K_d models vary for different turbid waters.

1) For moderately turbid waters, $K_d(\lambda) = K_d^{MT}(\lambda)$ was calculated using a and b_{bp} and b_{bw} (backscattering coefficient of pure water) [1], [68]

$$k_d^{MT}(\lambda) = (1 + k_0 \times \theta) \times a(\lambda) + \left(1 - \gamma \times \frac{b_{bw}(\lambda)}{b_b(\lambda)}\right) \times k_1 \times \left(1 - k_2 \times e^{-k_3 \times a(\lambda)}\right) \times b_b(\lambda) \quad (8)$$

where θ represents the sun zenith angle, b_b represents the backscattering coefficient (the sum of b_{bw} and b_{bp}), and k_{0-3} are 0.005, 4.26, 0.52, and 10.8, respectively.

2) For extremely turbid waters, $K_d(\lambda) = K_d^{ET}(\lambda)$ was calculated using a and θ [69]

$$k_d^{ET}(\lambda) = (1 + COS\theta) \times a(\lambda) + \left[\left(a(\lambda)^3 + 1\right)a(\lambda)\right]^{0.01}. \quad (9)$$

Since the parameter values (k_{0-3}) of (8) were originally derived from marine water, there may be differences when applied to inland waters. To address this, Zeng et al. [33] made modifications using measured data, while Simon and Shanmugam [69] constructed a K_d calculation model applied to turbid waters. To assess the differences generated by the K_d models, different K_d models were compared for Z_{SD} in moderately turbid and extremely turbid waters, and the detailed results were shown in Fig. S4 [33], [68], [69]. On the one hand, the model by Lee et al. [68] has the most accurate estimations in moderately turbid waters; on the other hand, the model by Simon and Shanmugam [69] demonstrates the highest accuracy in extremely turbid waters.

Finally, Z_{SD} was deriving using K_d and R_{rs}. Lee et al. [29] improved Z_{SD} retrieval by addressing certain limitations or mistakes in the classical model and proposed a novel theoretical model for remotely estimating Z_{SD}

$$Z_{SD} = \frac{1}{2.5\text{Min}\left(K_d^{tr}\right)} \times \ln\left(\frac{|0.14 - R_{rs}^{tr}|}{0.013}\right) \quad (10)$$

where K_d^{tr} is the spectral transparent window for K_d and the minimum values among 443, 488, 532, 555, and 665 nm are generally taken, R_{rs}^{tr} is the corresponding wavelength of R_{rs}.

IV. Results

A. Semianalytical Model of Z_{SD} Based on Simulated OHS Image

Fig. 8 illustrates a comparison between the OHS-simulated Z_{SD} and the measured Z_{SD}. Previous studies have indicated discrepancies between the actual values derived from model outcomes and the measured Z_{SD} [18]. These disparities are potentially caused by the QAA model's tendency to overestimate K_d in turbid conditions, resulting in an underestimated estimation of Z_{SD} [4]. To address this concern, a significant correlation between the two parameters has been identified [4], [18]. In Fig. 8, the red symbol represents the correction results for Z_{SD}, indicating that the consistency of the scaled $Z_{SD}(Z'_{SD})$ has been significantly improved compared with the original Z_{SD}, with a mean-MAPD of 10.89% and a mean-RMSD of 0.125 m, respectively. Z'_{SD} is calibrated as follows:

$$\text{Moderately turbid}: Z'_{SD} = 1.1697 Z_{SD} + 0.2184 \quad (11)$$
$$\text{Extremely turbid}: Z'_{SD} = 1.8445 Z_{SD} - 0.0691. \quad (12)$$

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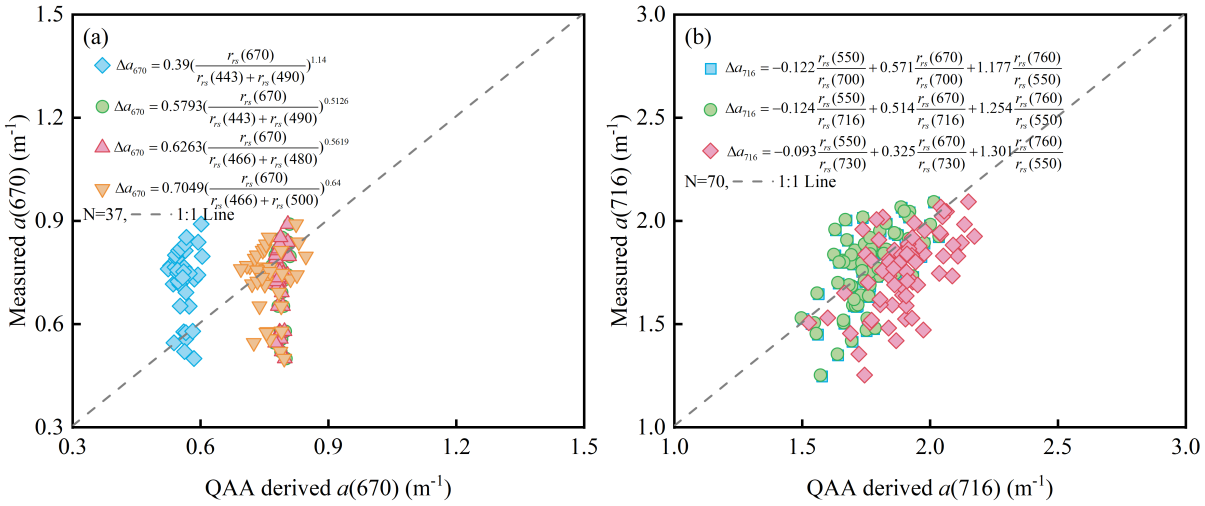


Fig. 5. Fitting curves of QAA-derived $a(\lambda)$ and measured $a(\lambda)$ with different bands. (a) QAA_{670}. (b) QAA_{716}.

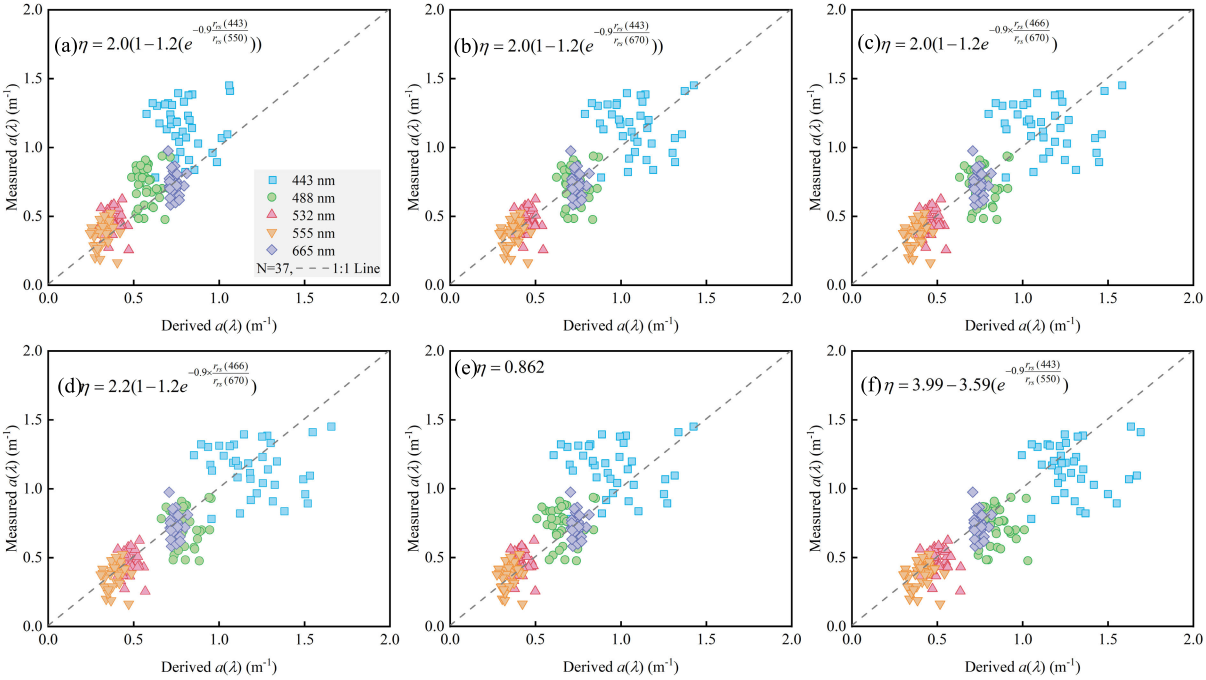


Fig. 6. Performance of QAA_{670} estimated $a(\lambda)$ under different η. (a) and (b) η obtained from Lee et al. [1]. (c) η using OHS band settings. (d) η adjusted from Rodrigues et al. [67]. (e) η obtained from Huang et al. [61]. (f) η obtained from Yang et al. [37].

B. Validation of OHS-Derived Z_{SD} With AC Models

Four AC models (6S, FLAASH, DOS, and QUAC) were compared to reduce the effect of the atmosphere on the OHS-derived Z_{SD}. The 88 uniformly distributed satellite-based Z_{SD} data synchronously collected from DCL and EHL (detailed classification in Fig. S5). As depicted in Fig. 9, the FLAASH model presents promising accuracy in moderately turbid and extremely turbid waters, resulting in an MAPD of 23.35% and RMSD of 0.223 m. Conversely, the 6S model can be used in extremely turbid waters but is inefficient in moderately turbid waters, obtaining an MAPD of 29.87% and RMSD of 0.304 m. The DOS and the QUAC models could not be applied to the OHS images, due to the poor accuracy (MAPD > 38%). These findings indicate that the FLAASH model can be effectively utilized for OHS monitoring of Z_{SD} in the turbid plateau waters and exhibiting satisfactory performance. The use of the FLAASH model for OHS images has been suggested in previous studies [23], [52].

C. Spatial–Temporal Patterns of Z_{SD}

Fig. 10 shows the Z_{SD} spatial–temporal patterns obtained using the novel semianalytical model. In the DCL region, Z_{SD} displayed significant spatial–temporal variability, with an increasing trend from the lake center toward the shore (mean Z_{SD} of 0.37 m). The lowest documented Z_{SD} of 0.26 m was observed in November 2020, while the highest documented Z_{SD} of 0.56 m was observed in February 2021. On the other hand, in the EHL region,

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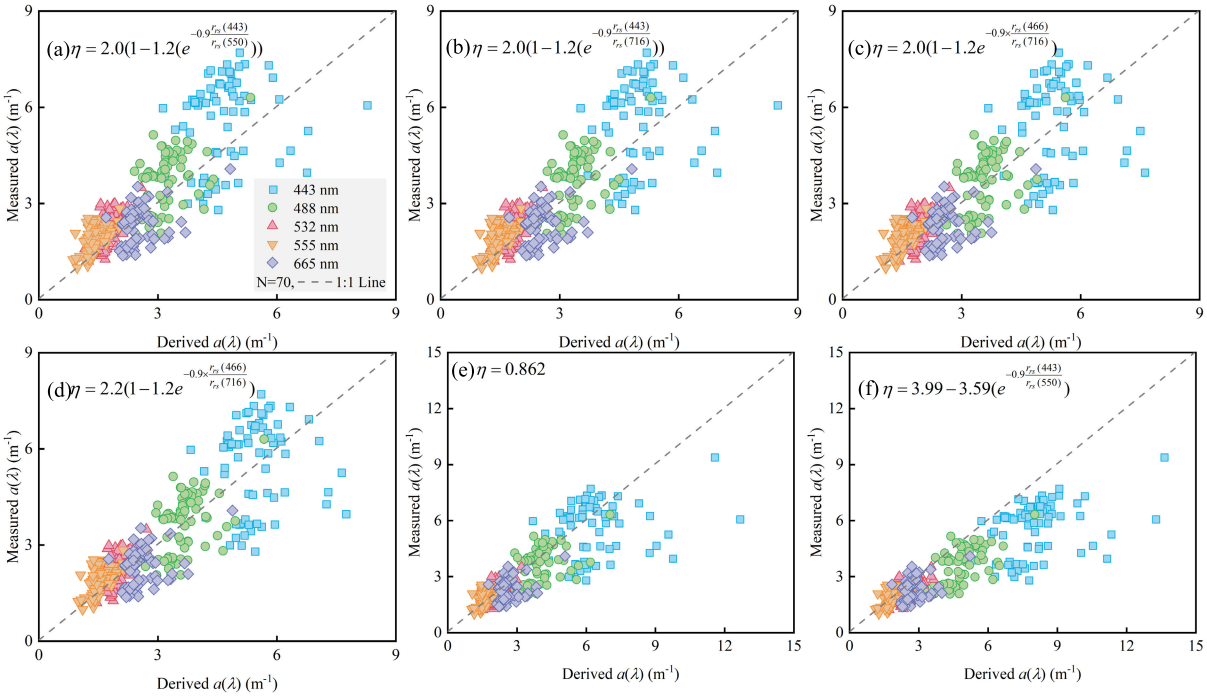


Fig. 7. Performance of QAA_{716} estimated $a(\lambda)$ under a different η. (a) and (b) η obtained from Lee et al. [1]. (c) η using OHS band settings. (d) η adjusted from Rodrigues et al. [67]. (e) η obtained from Huang et al. [61]. (f) η obtained from Yang et al. [37].

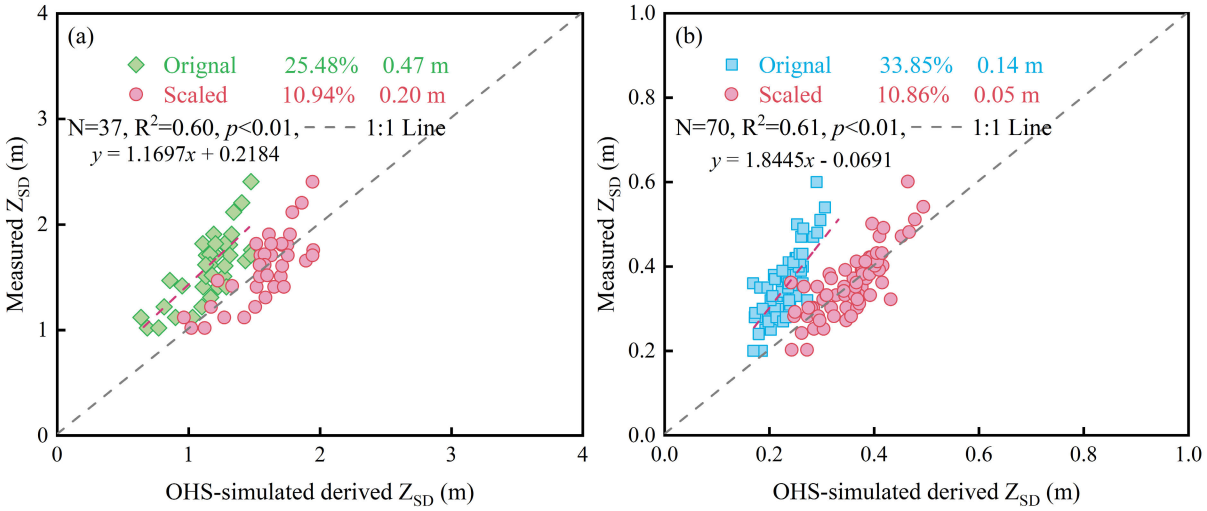


Fig. 8. Comparison with the original and scaled Z_{SD}. (a) Moderately turbid. (b) Extremely turbid.

Z_{SD} showed an observed spatial–temporal distribution, with a decreasing trend from north to south (mean Z_{SD} of 1.22 m). The lowest recorded Z_{SD} of 0.98 m was observed in December 2021, while the highest recorded Z_{SD} of 1.81 m was observed in December 2020. Previous studies have also identified spatial–temporal patterns in DCL and EHL [40], [70].

V. Discussion

A. Evaluation of Different QAA Models

To assess the QAA_{hybrid}model' estimation capability, it was compared with the QAA_{V6} model, as well as two improved QAA models introduced by Jiang et al. [8] (QAA_{J19}) and Liu et al. [36] (QAA_{L22}). Fig. 11(a)–(d) illustrates the validation results of different QAA model estimates a at 488, 555, and 665 nm [these all Z_{SD}-related bands, (10)]. The QAA_{hybrid} shows a well-accepted accuracy (with MAPD from 19.04% to 21.22% and RMSD from 0.38 to 0.67 m^{-1}) than other QAA models. In contrast, the QAA_{V6} displays a seriously underestimated phenomenon, with explicit biases across all bands, with MAPD exceeding 50% and RMSD exceeding 1.0 m^{-1}. The QAA_{J19} adjusts for deviations but still produces substantial overestimation in the turbid waters and overall underestimations in clear waters, with MAPD exceeding 30% and RMSD exceeding 0.50 m^{-1}. The QAA_{L22}

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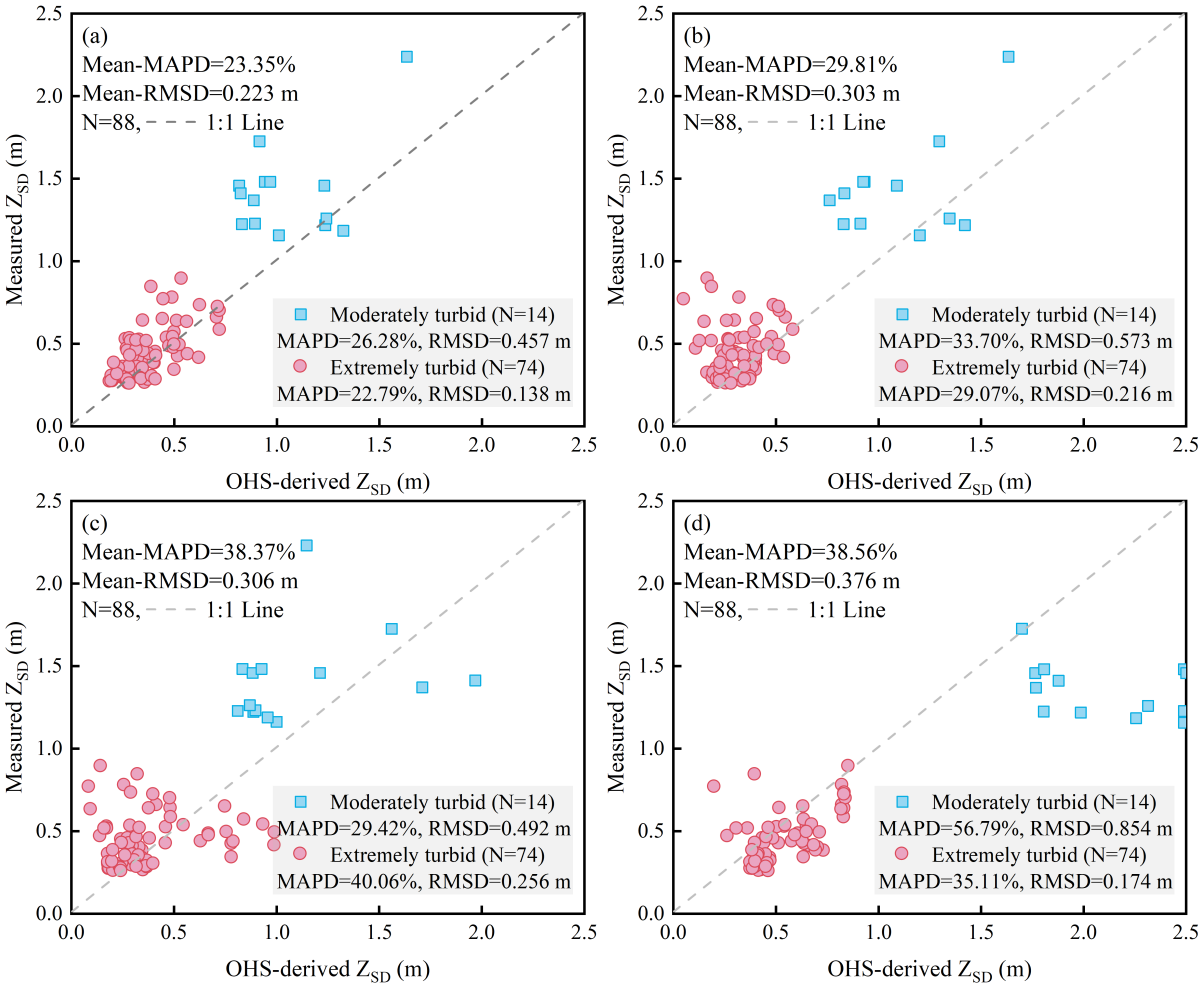


Fig. 9. Scatter plots of OHS-derived Z_{SD} and measured Z_{SD}. (a) FLAASH model. (b) 6S model. (c) QUAC model. (d) DOS model.

generally provides reliable results, but its estimations at longer wavelengths show slight deviations, with MAPD exceeding 29% and RMSD exceeding 0.50 m^{-1}. The results suggest that the QAA_{hybrid} model outperforms existing QAA models and can be effectively utilized for estimating $a(\lambda)$ with satisfactory performance.

Fig. 11(e)–(h) displays the error propagation of the QAA_{hybrid} model and other QAA models. The QAA_{hybrid} model selected 670 and 716 nm as the reference bands, resulting in the smallest errors propagated to the shortwave bands, with the average bias ranged from 4.25% to 18.28% for extremely turbid waters and from 3.05% to 3.65% for moderately turbid waters. In contrast, QAA_{V6} considered 670 nm as the reference band for both DCL and EHL, leading to the highest bias with an average bias of from 37.45% to 60.86%. QAA_{J19} utilized the maximum chlorophyll index (MCI) values to convert the reference bands (560 and 754 nm), which reduced error propagation to the shortwave bands, with an average bias of from 11.59% to 24.51% for the turbid waters, and from 15.28% to 19.53% for the clear waters. QAA_{L22} adopted 555 and 708 nm as the reference bands, which generally reduced error propagation to the shortwave bands but performed poorly in clear water as the water classifications. These results show that the reference band combination of 670 and 716 nm has a better performance and reduces the errors in the extrapolated bands.

B. Comparison With Existing Algorithms

The existing Z_{SD} estimation models available for the DCL and EHL regions have limitations in their applicability. For their evaluation, various models were simulated using measured data from DCL and EHL, with the detailed results for Table II and Figs. S6–S8.

When comparing the existing Z_{SD} models, the model using QAA_{hybrid} proposed in this study demonstrated the most promising performance. Among the empirical models, one-band and two-band models following the classification method of this study showed relatively satisfactory accuracy but weak correlation. Conversely, models following the classification of Lu et al. [71] exhibited notable bias, with a relatively high MAPD exceeding 37%. Regarding the semianalytical models, QAA_{V6} showed the poorest performance in estimating Z_{SD} (MAPD = 80.33% and RMSD = 0.430 m), due to the substantial underestimation performance of a (see Fig. 11).

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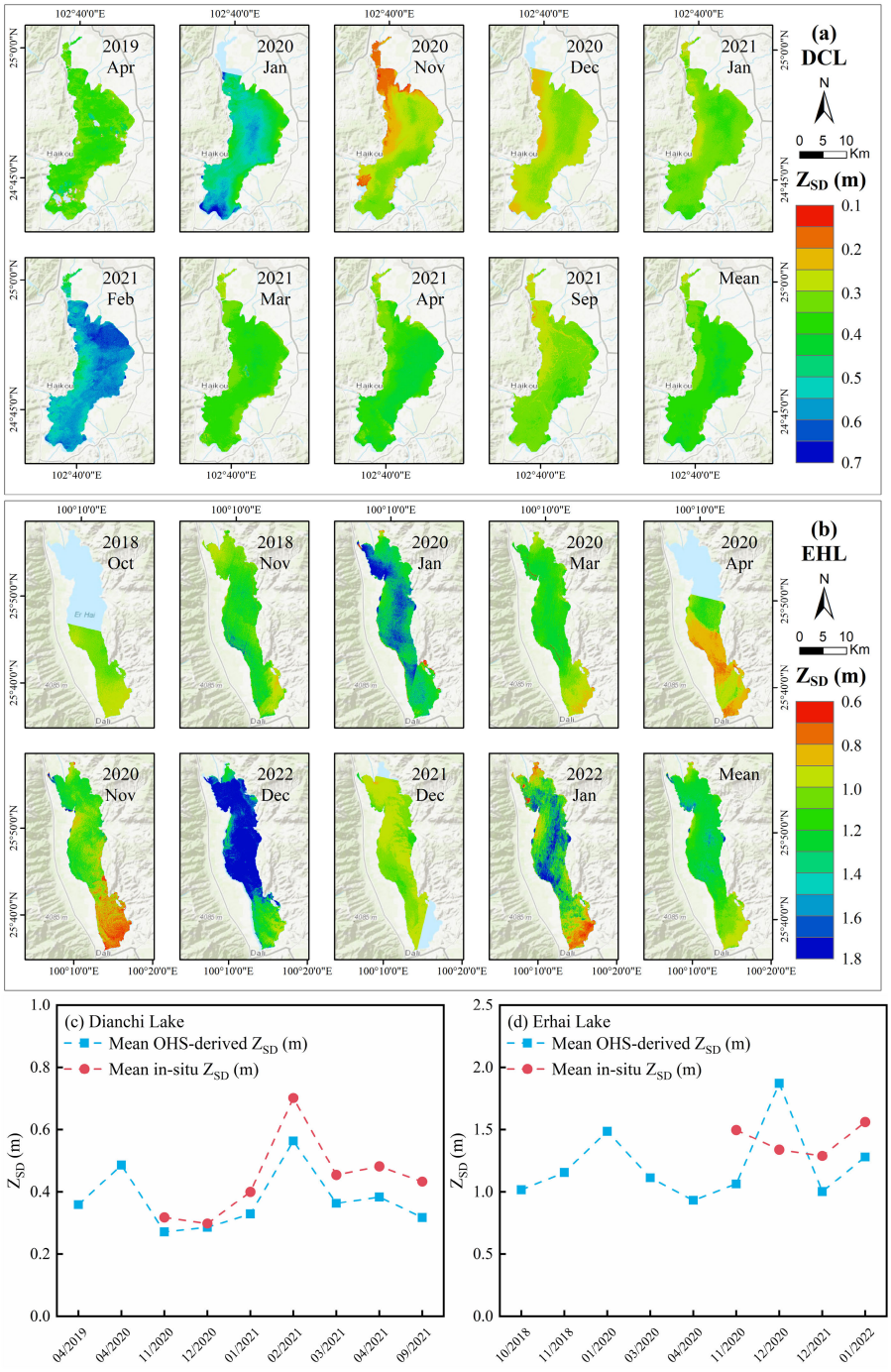


Fig. 10. OHS-derived Z_{SD} spatial–temporal patterns in DCL and EHL. (a) and (b) Z_{SD} spatial–temporal mapping. (c) and (d) Comparison of OHS-derived Z_{SD} with in situ Z_{SD}.

QAA_{R18}, which utilized 708 nm as the reference band, outperformed QAA_{V6} but was still inadequate for accurate Z_{SD} estimation (MAPD = 69.29% and RMSD = 0.555 m). Models such as QAA_{J19} and QAA_{Z20}, incorporating far near-infrared bands (754 and 745 nm, respectively), displayed similar performances. While QAA_{J19} employed MCI for water classification, QAA_{Z20} adjusted the K_d model. In addition, QAA_{L20}, which also choose 708 nm as the reference band, achieved improved accuracy compared to QAA_{R18} (MAPD = 39.36% and RMSD = 0.308 m), possibly due to the rationalization of water classification. On the other hand, SAM_{X23} demonstrated a certain ability to differentiate water types but failed to deliver satisfactory performance in estimating Z_{SD} (MAPD = 60.62% and RMSD = 0.644 m). The poor results stem from the fact that its design focuses on the ocean waters.

In conclusion, for the QAA models, a single reference band is insufficient to accurately estimate Z_{SD} in complex inland waters, such as DCL and EHL. Selecting 670 and 716 nm as reference bands could provide more accurate estimates in these regions.

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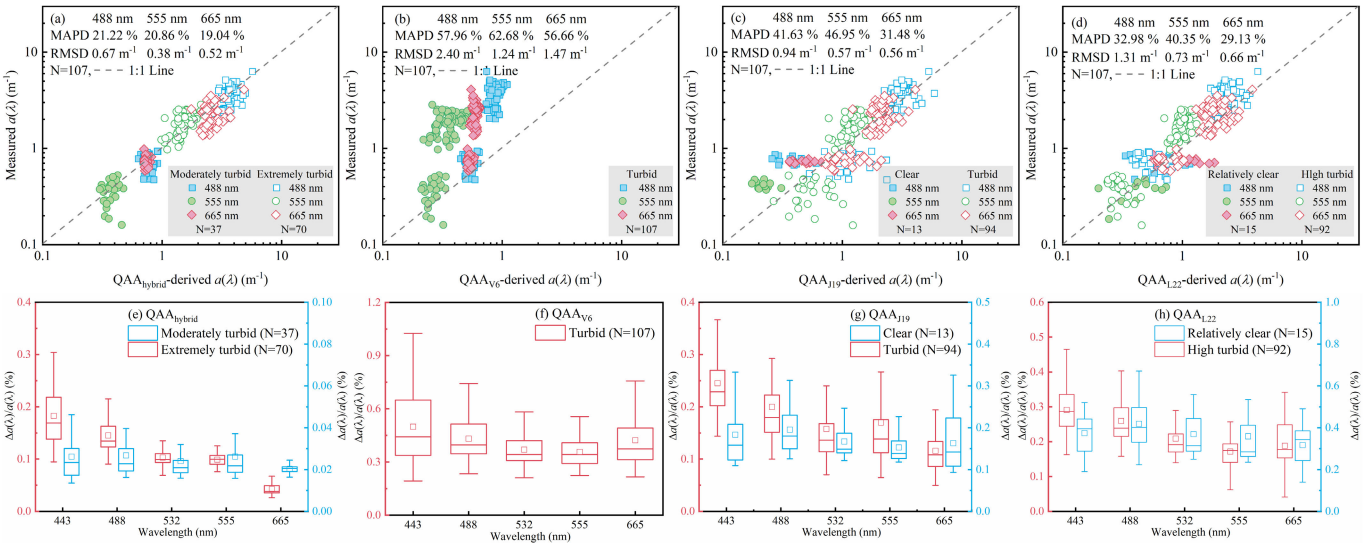


Fig. 11. Evaluation of different QAA models (QAA_{hybrid}, QAA_{V6}, QAA_{J19}, and QAA_{L22}). (a)–(d) Scatter plot of QAA-estimated a and measured a. (e)–(h) Extrapolation errors for QAA models.

TABLE I

STEPS OF SEMIANALYTICAL MODEL. STEPS WITH A GRAY BACKGROUND ARE THE MODIFIED PART OF THIS STUDY. U(λ) IS THE PARAMETER FROM LEE ET AL. [65]

Part	Step	Equations
Part I: QAA_{hybrid} model	1	$r_{rs}(\lambda)=R_{rs}(\lambda)/(0.52+1.7\times R_{rs}(\lambda))$
	2	$u(\lambda)=\dfrac{-g_0+\sqrt{g_0^{\,2}+4g_1\times r_{rs}(\lambda)}}{2g_1}$, $g_0=0.089, g_1=0.125$
	3	IF $f=R_{rs}(670)/R_{rs}(500)<0.70$: $a(\lambda_0)=a_w(670)+0.7049\left(\dfrac{r_{rs}(670)}{r_{rs}(466)+r_{rs}(500)}\right)^{0.64}$; ELSE $f=R_{rs}(670)/R_{rs}(500)\geq 0.70$: $a(716)=a_w(716)+\alpha\dfrac{r_{rs}(550)}{r_{rs}(700)}+\beta\dfrac{r_{rs}(670)}{r_{rs}(700)}+\gamma\dfrac{r_{rs}(760)}{r_{rs}(550)}$, $\alpha=-0.122,\beta=0.571,\gamma=1.177$
	4	$b_{bp}(\lambda_0)=\dfrac{u(\lambda_0)\times a(\lambda_0)}{1-u(\lambda_0)}-b_{bw}(\lambda_0)$
	5	$\eta=2.0\times[1-1.2\times e^{(-0.9\times\frac{r_{rs}(466)}{r_{rs}(670)})}]$ $\quad$ $\eta=2.2\times[1-1.2\times e^{(-0.9\times\frac{r_{rs}(466)}{r_{rs}(716)})}]$
	6	$b_{bp}(\lambda)=b_{bp}(\lambda_0)\times\left(\dfrac{\lambda_0}{\lambda}\right)^{\eta}$
	7	$a(\lambda)=\dfrac{(1-u(\lambda))\times(b_{bw}(\lambda)+b_{bp}(\lambda))}{u(\lambda)}$
Part II: Z_{SD} estimated model	8	$k_d^{MT}(\lambda)=(1+k_0\times\theta)\times a(\lambda)+(1-\gamma\times\dfrac{b_{bw}(\lambda)}{b_b(\lambda)})\times k_1\times(1-k_2\times e^{-k_3\times a(\lambda)})\times b_b(\lambda)$ $\quad$ $k_d^{ET}(\lambda)=(1+COS\theta)\times a(\lambda)+\left[(a(\lambda)^3+1)a(\lambda)\right]^{0.01}$
	9	$Z_{SD}=\dfrac{1}{2.5Min(K_d^{tr})}\times\ln\left(\dfrac{\lvert 0.14-R_{rs}^{tr}\rvert}{0.013}\right)$
	10	$Z_{SD}^{MT}=1.1697Z_{SD}+0.2184$ $\quad$ $Z_{SD}^{ET}=1.8445Z_{SD}-0.0691$

C. Sensitivity of Semianalytical Models

Fig. 12 displays the variations in the semianalytical model under different random error conditions (from −50% to 50%). Fig. 12(a) shows the comparison of the sensitivity of water classification using $R_{rs}(500)$ and $R_{rs}(670)$. Within the error range of 25%–50%, the MAPD of $R_{rs}(500)$ increases rapidly, while for $R_{rs}(670)$, the MAPD increases extremely within the error range from −50% to −25%. In contrast, within an

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TABLE II

COMPARISON OF THE EXISTING AND PROPOSED MODELS FOR ESTIMATING Z_{SD} IN THE DCL AND EHL. BI IS THE OHS SIMULATED *I* TH BAND. QAA_{XX} IS THE MODEL FROM XX, AND THE SAM IS SEMIANALYTICAL MODEL FORM XIANG ET AL. [63]. AHSI IS THE ADVANCED HYPERSPECTRAL IMAGER. TYPE 1 AND TYPE 2 ARE THE CLASSIFICATION FROM LU ET AL. [71]

Type	Model	Sensors	Model Form	R^2	MAPD (%)	RMSD (m)
The empirical model	Red [72]	MODIS	Mt: $Z_{SD}=0.4104e^{-5.029*B14}$	0.03	25.32	0.435
			Et: $Z_{SD}=2.0979e^{-58.92*B14}$	0.37	27.14	0.093
	Green/Red [73]	OLI	Mt: $Z_{SD}=0.6462e^{-0.841*(B13/B7)}$	0.14	28.55	0.436
			Et: $Z_{SD}=0.6769e^{-2.2433*(B13/B7)}$	0.14	20.02	0.073
	Optical classification [71]	OLI	Type 1: $Z_{SD}=1.9755*(B1/B13)-0.7798$	0.76	37.74	0.367
			Type 2: $Z_{SD}=3.1967*(B1/B13-B1/B7)-0.2673$	0.96	40.41	0.296
The semi-analytical model	QAA_{V6} [1]	OLI	Step 1: $R_{rs}(665)$ →Water type Step 2: QAA_{V6} →a(λ)&b(λ) Step 3: $a(\lambda)$&b(λ) →$K_d(\lambda)$ Step 4: $K_d(\lambda) \rightarrow Z_{SD}$		80.33	0.430
	QAA_{R18} [74]	OLCI	Step 1: QAA_{R18} → a(λ)&b(λ) Step 2: $a(\lambda)$&b(λ) →$K_d(\lambda)$ Step 3: $K_d(\lambda) \rightarrow Z_{SD}$		69.29	0.555
	QAA_{J19} [8]	MERIS	Step 1: MCI →Water type Step 2: QAA_{J19} →a(λ)&b(λ) Step 3: $a(\lambda)$&b(λ) →$K_d(\lambda)$ Step 4: $K_d(\lambda) \rightarrow Z_{SD}$		51.59	0.686
	QAA_{Z20} [33]	GOCI	Step 1: QAA_{Z20} → a(λ)&b(λ) Step 2: $a(\lambda)$&b(λ) → $K_d(745)$ Step 3: $K_d(745)$ →$K_d(555)$ Step 4: $K_d(555) \rightarrow Z_{SD}$		51.91	0.742
	QAA_{L22} [36]	AHSI	Step 1: $R_{rs}(665)/R_{rs}(510)$ →Water type Step 2: QAA_{L22} →a(λ)&b(λ) Step 3: $a(\lambda)$&b(λ) →$K_d(\lambda)$ Step 4: $K_d(\lambda) \rightarrow Z_{SD}$		39.36	0.308
	SAM_{X23} [63]	Multiple sensors	Step 1: $R_{rs}(670)/R_{rs}(490)$ → Water type Step 2: Water type → $K_d(\lambda)$ Step 3: $K_d(\lambda)$ →Z_{SD}		60.62	0.644
This study	QAA_{hybrid}	OHS	Step 1: $R_{rs}(670)/R_{rs}(500)$ →Water type Step 2: QAA_{hybrid} →a(λ)&b(λ) Step 3: $a(\lambda)$&b(λ) →$K_d(\lambda)$ Step 4: $K_d(\lambda) \rightarrow Z_{SD}$ Step 5: $Z_{SD}' = aZ_{SD,}+b$		10.89	0.125

error range (±20%), the MAPD of $R_{rs}(500)$ and $R_{rs}(670)$ increase slowly. Fig. 12(b) and (c) shows the comparison of the sensitivity of the key parameters in the Z_{SD} semi-analytical model: $a(\lambda_0)$, $K_{d,}$ and original Z_{SD}. Among these parameters, original Z_{SD} shows a nearly linear relationship with error, whereas K_d and $a(\lambda_0)$ demonstrate pronounced sensitivity with the error range from −50% to 0%, with their MAPD increasing rapidly from −50% to −30%. For moderately turbid water, the sensitivity follows the sequence: $K_d = a(\lambda_0) >$ original Z_{SD}, while for extremely turbid water and the sensitivity shifts to: $K_d > a(\lambda_0) >$ original Z_{SD}. This discrepancy may be attributed to variations in K_d models across different water types. Specifically, the K_d^{ET} value is low, the relationship between K_d and Z_{SD} is approximately inversely proportional (10), leading to dramatic changes in MAPD within the error range of −50% to −30%.

These findings highlight the critical role of the K_d model and water classification methods in accurately retrieving Z_{SD}. Moreover, the method proposed in this study demonstrates stable performance within a certain error range (±20%), emphasizing the importance of adopting more refined strategies to improve model reliability in complex waters.

D. Applicability and Limitations of OHS Image

Fig. 13 shows the comparison of NE_{ZSD} of different hyperspectral satellite images using QAA_{hybrid} [23], [75]. The NE_{ZSD} derived from four images increased with the solar zenith angle (θ), the performance of these images was affected by θ. [23], [76]. The relationship between NE_{ZSD} for the four images remained consistent: OLCI had the lowest NE_{ZSD} (0.128 m), followed by MODIS (0.217 *m*) and OHS (0.268 m), while

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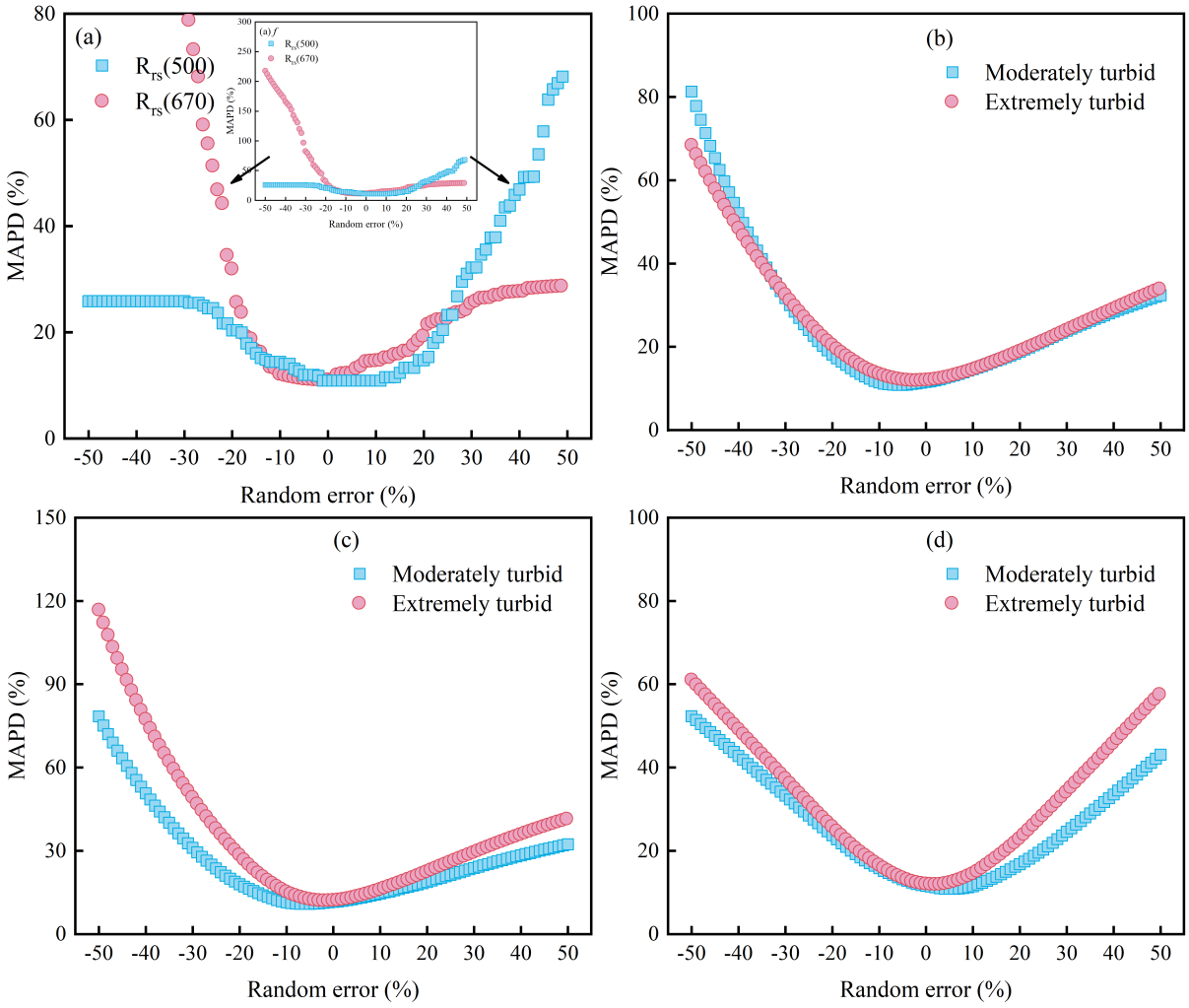


Fig. 12. Sensitivity analysis of the Z_{SD} semianalytical model. (a) f. (b) $a(\lambda_0)$. (c) K_d. (d) Original Z_{SD}.

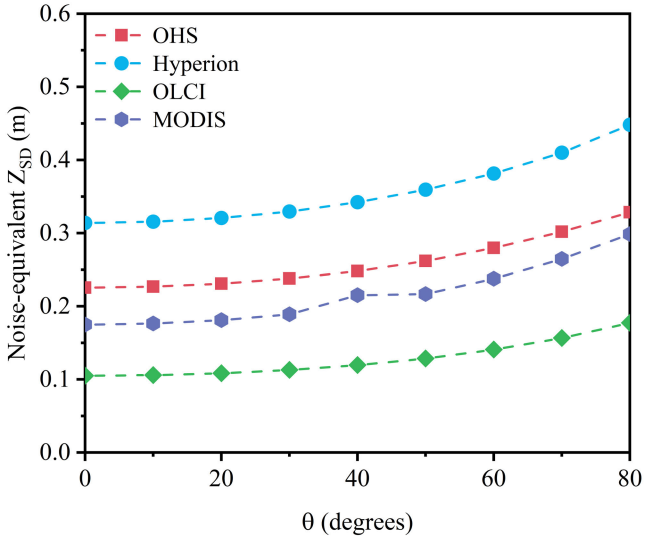


Fig. 13. NE_{ZSD} for different sensors using QAA_{hybrid}.

Hyperion had the highest NE_{ZSD} (0.358 m). In conclusion, the radiometric performance of OHS images was found to be satisfactory, with low noise impact achieved through the application of the newly proposed model in this study.

OHS imagery not only exhibited remarkable potential in this study but also faced certain uncertainties and limitations. First, the application of this study focused on DCL and EHL, indicating that further exploration is needed for other turbid plateau lakes, such as Xingyun Lake [77]. Second, to assess the stability and expandability of OHS imagery, necessitating further long-term monitoring [39]. In addition, while this study compared different AC models, it is essential to explore other AC models as well, such as Acolite [43], [53], [78].

VI. Conclusion

An improved QAA_{hybrid} was employed to estimate Z_{SD} in turbid plateau waters. This model utilized optical classification to estimate IOPs. Subsequently, a linearly corrected for Z_{SD} was applied to align the model with in situ measurements. When compared with the three existing QAA models (QAA_{V6}, QAA_{J19}, and QAA_{L22}), QAA_{hybrid} was well-accepted for its high accuracy in $a(\lambda)$ estimation, with MAPD reduced to 19.04%. By using a combination of 670 and 716 nm as reference bands, the QAA_{hybrid} model produced the lowest

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error among the extrapolated bands. The newly introduced Z_{SD} semianalytical model demonstrated better performance compared with existing models through rigorous analysis (MAPD = 10.89%), with robust results observed in OHS images (MAPD = 23.42%). The OHS imagery yielded relatively moderate NE_{ZSD} values, ranging from 0.23 to 0.33 m, effectively decreasing the noise and improving Z_{SD} retrieval. Notable variations in spatial–temporal Z_{SD} patterns were identified between DCL and EHL. The central region of DCL appeared more transparent than its edges, while the northern region of EHL exhibited greater transparency compared with the southern part. The algorithm's application to a wider range of waters and its ability to handle extreme conditions, addressing the limitations of a single model in complex aquatic environments and providing valuable support for water environmental management and sustainable development efforts.

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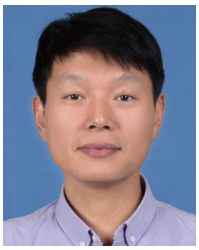
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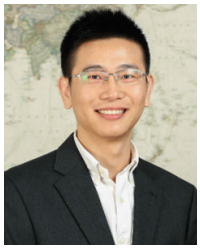
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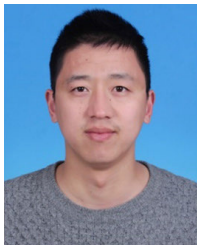
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